

# Pranshu Patel

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## Summary

Backend-focused Java developer specializing in Spring Boot — JWT authentication, pessimistic locking, Redis soft locks, Resilience4j circuit breakers, and idempotency enforcement. Currently building an LLM-powered SQL analytics agent using Spring AI with pgvector schema retrieval and parser-level query safety. Deployed two production-style backends as modular monoliths with **90%+ service-layer test coverage**, Flyway-managed schemas, and Docker cloud deployment.

## Education

**L.D. College of Engineering (LDCE)**

Bachelor of Engineering in Information Technology — CGPA: 8.8/10

July 2023 – June 2027

Ahmedabad, Gujarat

## Technical Skills

**Languages:** Java, SQL

**Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST API, JDBC

**AI & Data:** Spring AI, LLM Integration (Groq / Ollama), pgvector Embedding Search, JSQParser

**Concepts:** Modular Monolith Architecture, JWT Auth, RBAC, DTO Pattern, Pessimistic Locking, Idempotency, Transaction Management, Pagination & Sorting

**Testing & DevOps:** JUnit 5, Mockito, Docker, Docker Compose, Git, Maven, Postman, Swagger/OpenAPI, Flyway, Resilience4j

**Databases:** MySQL, PostgreSQL, Redis

## Projects

**QuerySense — Intelligent SQL Analytics Agent** [\[GitHub\]](#) — Spring AI, PostgreSQL, pgvector, Redis, Docker *July 2026 – Present*

- Building an **AI agent that converts natural-language business questions into SQL** using **Spring AI** with LLMs (Llama 3.3 via Groq); grounds query generation on database schema context retrieved through **pgvector embedding similarity search**.
- Enforced **SQL safety for LLM-generated queries** via **JSQParser** validation and a dedicated **read-only analytics datasource**, isolating AI-executed queries from the application database.
- Orchestrated a multi-service stack (Spring Boot, dual PostgreSQL, **Redis**, **Ollama** embedding server) via **Docker Compose** with health-checked startup dependencies.

**Ticket Booking System** [\[Live Demo\]](#) — Spring Boot, PostgreSQL, Redis, Resilience4j, Flyway *March 2026 – June 2026*

- Implemented a **reserve-pay-confirm** booking flow using separate database transactions; **PESSIMISTIC\_WRITE** locks protect seat state during reservation and confirmation while payment executes outside the seat-lock transaction.
- Implemented **Redis seat holds** using atomic **SET NX PX** with a 5-minute TTL; Redis acts as an optimization layer while PostgreSQL row locking, the **RESERVED** state, and a **UNIQUE seat constraint** enforce booking correctness.
- Implemented **payment idempotency** using Redis as a fast cache backed by a PostgreSQL **payments** table with a **UNIQUE payment idempotency key**; duplicate requests resolve to the existing payment record instead of executing the payment again.
- Added a scheduled **reservation sweeper** to reclaim expired RESERVED seats and configured **Resilience4j Retry + Circuit Breaker** for payment failures; validated locking, concurrency, idempotency, and failure handling with **109 automated tests**; deployed on Render with Neon PostgreSQL and Upstash Redis.

**Secure Engineer Management API** [\[Live Demo\]](#) — Spring Boot, PostgreSQL, Docker

*Nov 2025 – Feb 2026*

- Built a Spring Boot REST API with **10+ secured endpoints** using layered architecture (Controller → Service → Repository); implemented **DTO layer** to eliminate sensitive field exposure; added dynamic filtering via **Spring Data JPA** Specifications with server-side **pagination & sorting**.
- Secured all endpoints with **JWT authentication** and **RBAC** via Spring Security; enforced ADMIN/USER access control at the method level using **@PreAuthorize** annotations.
- Achieved **90%+ service-layer test coverage** using JUnit 5 and Mockito with mocked dependencies; wrote controller-level integration tests with MockMvc.
- Implemented **centralized exception handling** via **@RestControllerAdvice** with **7 custom exception types** mapped to HTTP error responses (400/401/403/404/409); documented all endpoints with **Swagger/OpenAPI** and deployed via **Docker** on Render.

## Achievements

**Data Structures & Algorithms** — Solved **300+ problems** across LeetCode and GeeksforGeeks; ranked **44/299** across the institute on GeeksforGeeks.

**McKinsey Forward Program** — Completed McKinsey & Company's professional development program focused on problem solving and leadership. (Apr 2026 – Jun 2026)

## Certifications

**Introduction to Generative AI** — Google (Badge) | **Java Certification Test** — KnowledgeGate